

MASTER'S COMPREHENSIVE EXAMINATION

Part II Applied (90 minutes) December 7th, 2013

Instructions: Attempt any three out of the following four questions on both pages. Mention which three you would like to have graded. Open book and open notes.

1. Assume that birthdays are iid distributed across 365 days of the year (exclude Feb 29th on leap year). We have two groups of people

G1 – Has 10 Persons

G2 – Has 10 different persons

- A. What is the probability that no two persons in G1 were born on the same day of the year?
- B. Assume that no two persons in G1 have the same birthdate, then what is the probability that none of the persons in G2 were born on the same day as a person in G1?
- C. Assume again that no two persons in G1 have the same birthdate, what is the probability that none of the 20 persons in G1 and G2 combined have the same birthdate?

2. A linear regression model of Y = EGFR Kidney Function is fit with X = BUN (blood Urine Nitrogen) as the predictor to n = 200 persons and produces the following results

	Point Estimate	Std-Dev	
Intercept	$\hat{a} = 116$	1.15	$\text{Cov}(\hat{a}, \hat{b}) = -0.05$
BUN	$\hat{b} = -2.00$	0.10	$\sigma^2 = \text{Var}(Y X) = 3$

Assume normality with $Y = a + bX + \varepsilon$ where $\varepsilon \sim N(0, \sigma^2)$ and all observations are iid

- A. What is a 95% confidence interval for the Expected Value of EGFR if BUN = 13
- B. If George has BUN = 16 and Sam has BUN = 13 what is a 95% confidence interval for [George's EGFR – Sam's EFGR]

3. A nationwide dental chain is studying the variability in the time it takes a dental hygienist to clean a patient's teeth. They draw a random sample of 6 hygienists. For each hygienist they pick a random sample of 4 patients, and record the time to clean each patient's teeth. (The study involves a total of 24 patients.) The following table summarizes the mean and standard deviation of cleaning times for the patients for each hygienist

Hygienist	Patient time in minutes		
	Mean	standard deviation	number of patients
1	48	5	4
2	52	4	4
3	43	4	4
4	44	6	4
5	56	5	4
6	57	6	4

For the 24 patients the mean time is 50 minutes, and the standard deviation is 7.14447 minutes.

Write out ANOVA table showing source of variability, degrees of freedom, Sums of Squares, MS, and EMS. Estimate the components of variance

4. Soybean yields in a particular region are known to average 43 bushels per acre, with a standard deviation of 4 bushels. A researcher has developed a new variety of soybeans to be used in the region, that the researcher claims will increase yields to an average of 46 bushels per acre, with the same standard deviation of 4 bushels.

The researcher wants to take a random sample of n acres in the region, plant the new variety on each acre. The researcher's claim will be supported (at the one sided 0.05 level of significance) if the sample mean yield is large enough. The researcher wants the test to have a 90% chance of proving her claim (given that it is true).

How big should n be? Given that value of n , how big does the sample mean yield have to be to prove her claim.